



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

FREQUENCY ANALYSIS OF SIGNALS & SYSTEMS

- Understand the frequency response of LTI systems
- Know the ways to show the frequency response of LTI systems
- Understand the relationship between magnitude and phase

1. Frequency Response of LTI Systems
2. System Functions For Systems Characterized by Linear Constant-coefficient Difference equations
3. Frequency Response for Rational System Functions
4. Relationship Between Magnitude and Phase
5. All-Pass and Minimum-phase system
6. Linear Systems with Generalized Linear Phase

◆ **Frequency response** $H(e^{j\omega})$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

◆ **Magnitude response (gain)** $|H(e^{j\omega})|$

$$|Y(e^{j\omega})| = |H(e^{j\omega})| \cdot |X(e^{j\omega})|$$

change

deleterious
signal

system

distortions

◆ **Phase response (phase shift)** $\angle H(e^{j\omega})$

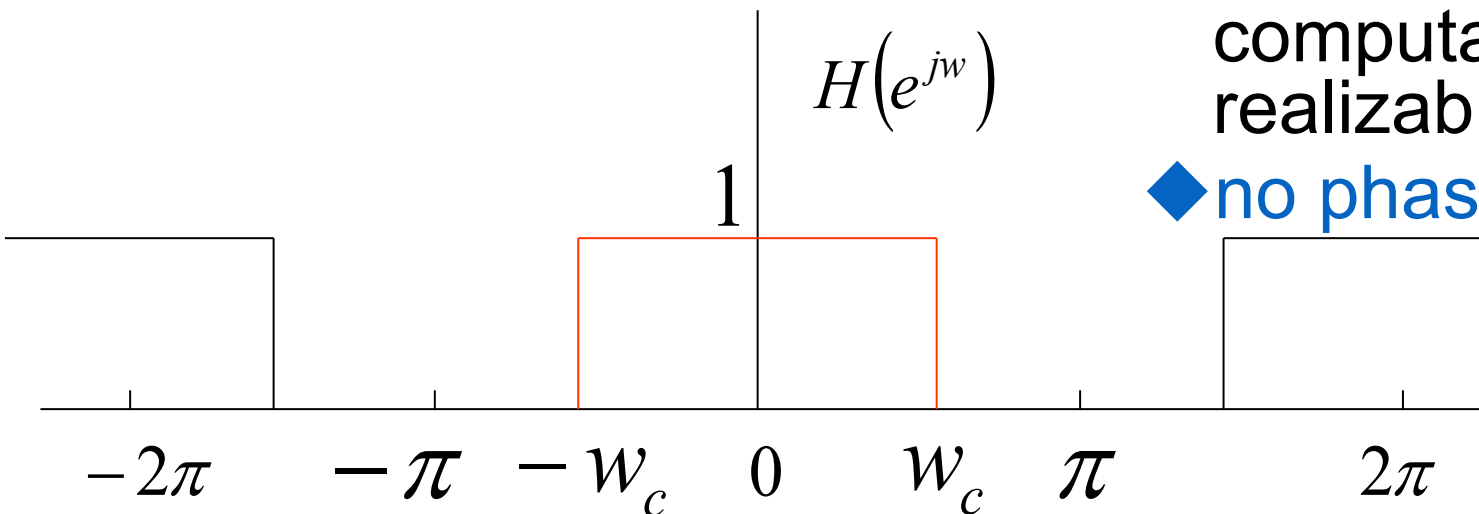
$$\angle Y(e^{j\omega}) = \angle H(e^{j\omega}) + \angle X(e^{j\omega})$$

- Ideal lowpass filter

$$H_{lp}(e^{j\omega}) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases}$$

$$h_{lp}(n) = \frac{\sin \omega_c n}{\pi n}, \quad -\infty < n < \infty$$

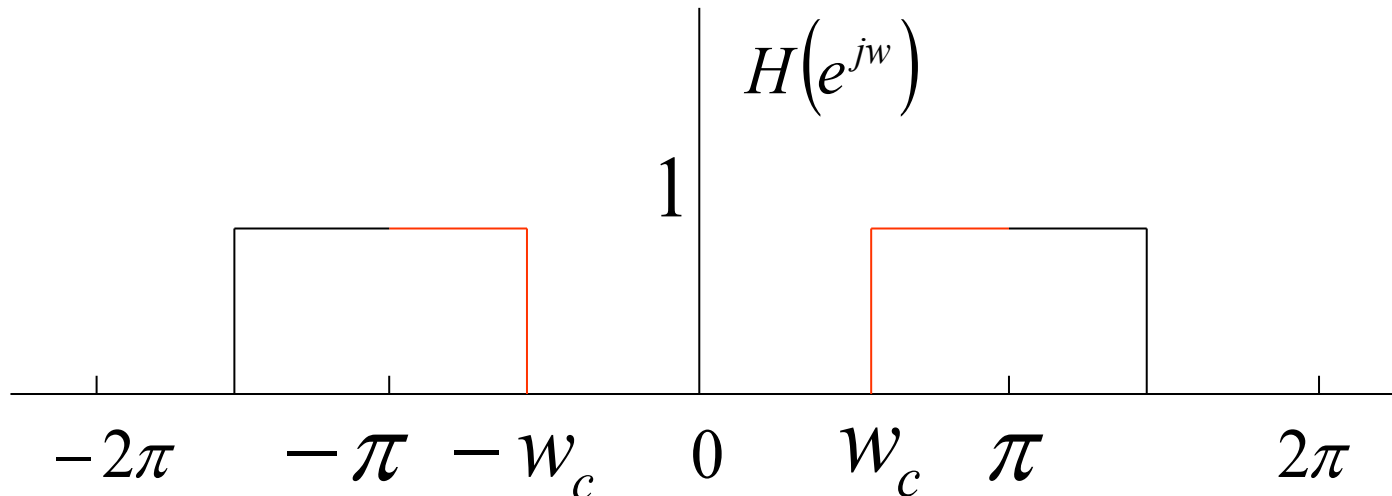
- ◆ Noncausal, not computationally realizable
- ◆ no phase distortion



- Ideal highpass filter

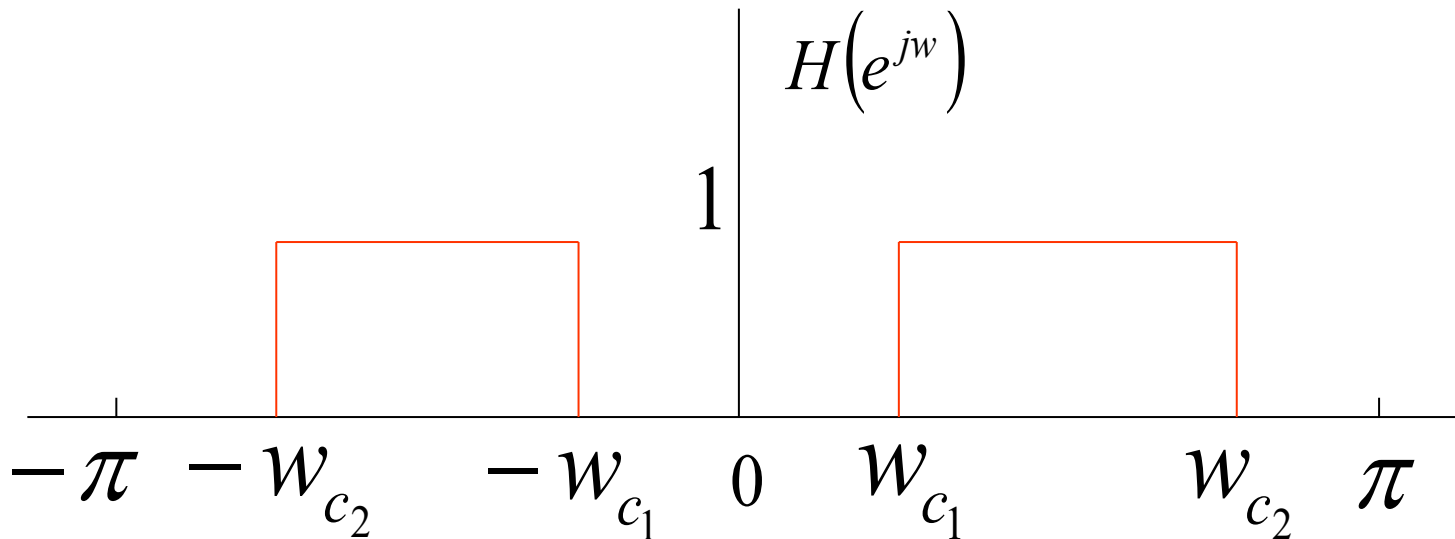
$$H_{hp} \left(e^{j\omega} \right) = \begin{cases} 0, & |\omega| < \omega_c \\ 1, & \omega_c < |\omega| \leq \pi \end{cases}$$

$$h_{hp} (n) = \delta[n] - \frac{\sin \omega_c n}{\pi n}, \quad -\infty < n < \infty$$



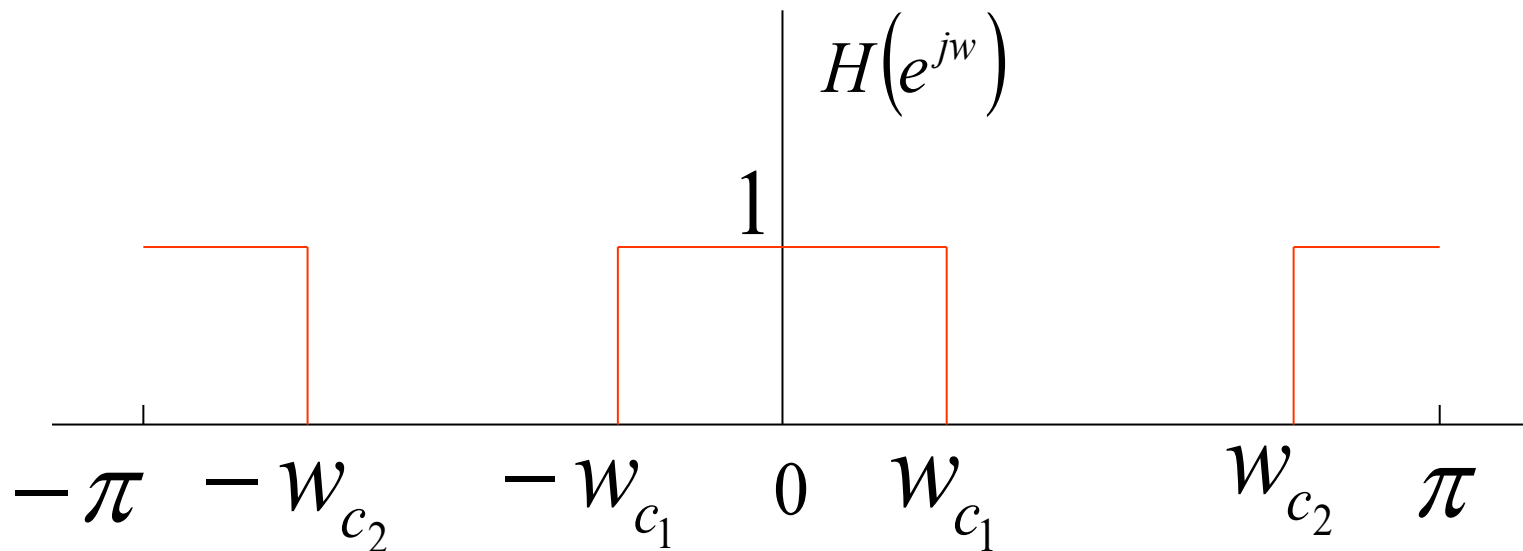
- Ideal bandpass filter

$$H_{bp}(e^{jw}) = \begin{cases} 1, & w_{c_1} < |w| < w_{c_2} \\ 0, & \text{others} \end{cases}$$



- Ideal bandstop filter

$$H_{bs}(e^{jw}) = \begin{cases} 0, & w_{c_1} < |w| < w_{c_2} \\ 1, & \text{others} \end{cases}$$



1. Frequency Response of LTI Systems
2. System Functions For Systems Characterized by Linear Constant-coefficient Difference equations
3. Frequency Response for Rational System Functions
4. Relationship Between Magnitude and Phase
5. All-Pass and Minimum-phase system
6. Linear Systems with Generalized Linear Phase

- Linear Constant-coefficient Difference equation

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

$$\sum_{k=0}^N a_k z^{-k} Y(z) = \sum_{k=0}^M b_k z^{-k} X(z)$$

$$\left(\sum_{k=0}^N a_k z^{-k} \right) Y(z) = \left(\sum_{k=0}^M b_k z^{-k} \right) X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \left(\frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k z^{-1})}{\prod_{k=1}^N (1 - d_k z^{-1})}$$

$$1 - c_k z^{-1} = 0 \quad \Rightarrow \quad z = c_k : \text{zero}$$

$$1 - d_k z^{-1} = 0 \quad \Rightarrow \quad z = d_k : \text{pole}$$

- Example: find **difference equation** for second-order System function

Solution:

$$H(z) = \frac{(1 + z^{-1})^2}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{3}{4}z^{-1}\right)} = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

$$\left(1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}\right)Y(z) = (1 + 2z^{-1} + z^{-2})X(z)$$

$$y[n] + \frac{1}{4}y[n-1] - \frac{3}{8}y[n-2] = x[n] + 2x[n-1] + x[n-2]$$

•Causal

- ◆ Impulse response $h[n]$ must be right-sided sequence.

$$h[n] = 0, \text{ for } n < 0$$

- ◆ The region of convergence of $H(z)$ must be outside the outermost pole.

- Stability

- The impulse response must be absolutely summable

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

- ROC of $H(z)$ includes the unit circle

$$\sum_{n=-\infty}^{\infty} |h[n] z^{-n}| < \infty \quad \text{for} \quad |z| = 1$$

- Example: Determine the ROC, Stability and causality for LTI system

$$y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n]$$

Solution:

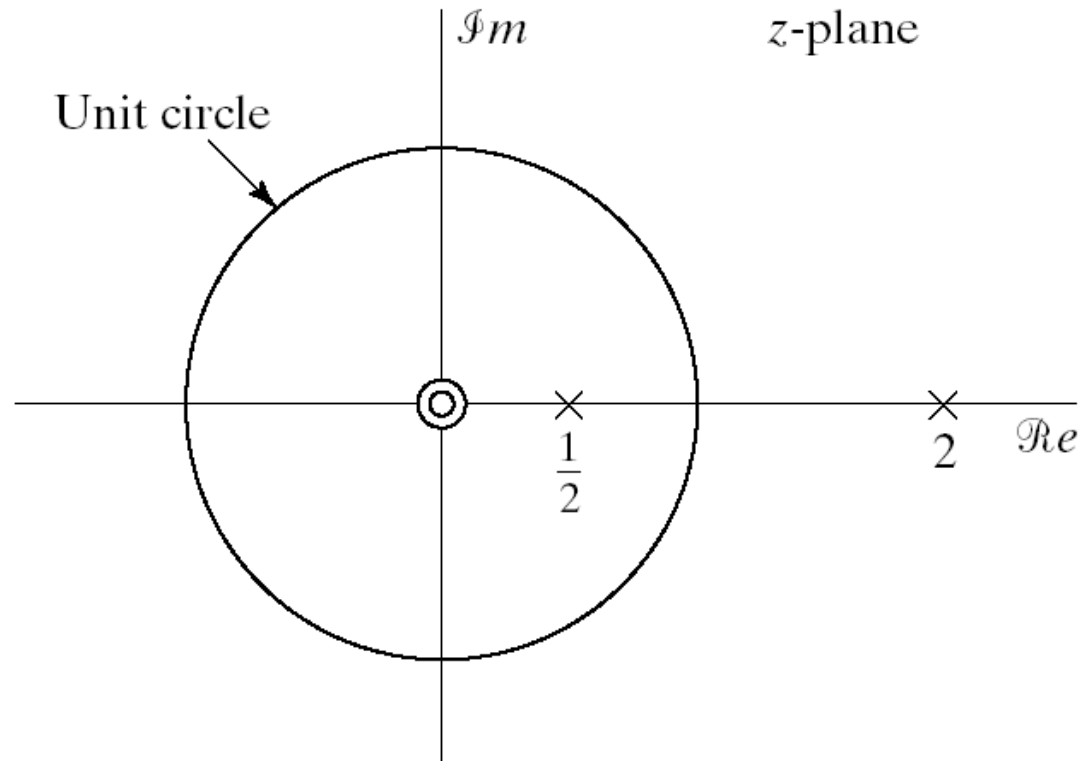
$$H(z) = \frac{1}{1 - \frac{5}{2}z^{-1} + z^{-2}} = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - 2z^{-1})}$$

◆ poles: 1/2, 2; zeros(two): 0

$|z| > 2 : \text{causal, not stable}$

$\frac{1}{2} < |z| < 2 :$

$\text{stable, not causal}$

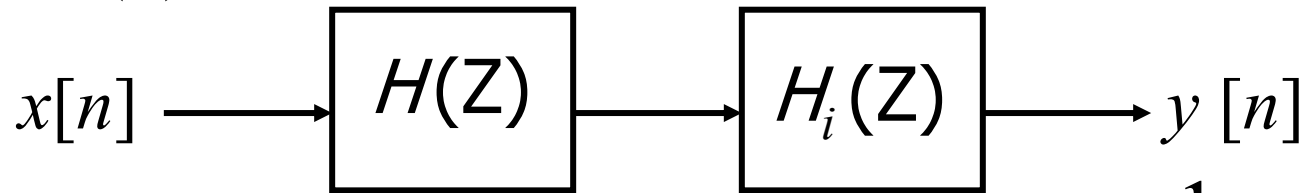


$|z| < \frac{1}{2} : \text{not causal, not stable}$

- **Causal**: ROC must be outside the outermost pole
- **Stable**: ROC includes the unit circle
- **Causal and stable**: all the poles of the system function are inside the unit circle.

• Inverse Systems

- ◆ For a LTI system $H(z)$, the inverse system $H_i(z)$ cascaded with $H(z)$ satisfies:



$$G(z) = H(z)H_i(z) = 1 \quad H_i(z) = \frac{1}{H(z)}$$

- ◆ Frequency response $H_i(e^{j\omega}) = \frac{1}{H(e^{j\omega})}$

- ◆ Time domain: $g[n] = h[n] * h_i[n] = \delta[n]$

$$H_i(z) = \frac{1}{H(z)}$$

$$H(z) = \left(\frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k z^{-1})}{\prod_{k=1}^N (1 - d_k z^{-1})} \quad H_i(z) = \left(\frac{a_0}{b_0} \right) \frac{\prod_{k=1}^M (1 - d_k z^{-1})}{\prod_{k=1}^N (1 - c_k z^{-1})}$$

◆ ROC of $H(z)$ and ROC of $H_i(z)$ must overlap.

- Example: analysis Inverse System for First-Order System

$$H(z) = \frac{1 - 0.5z^{-1}}{1 - 0.9z^{-1}} \quad ROC: |z| > 0.9$$

Solution:

$$H_i(z) = \frac{1 - 0.9z^{-1}}{1 - 0.5z^{-1}} \quad ROC: |z| > 0.5$$

$H_i(z)$: *stable and causal*

$$h_i[n] = (0.5)^n u[n] - 0.9(0.5)^{n-1} u[n-1]$$

1. Frequency Response of LTI Systems
2. System Functions For Systems Characterized by Linear Constant-coefficient Difference equations
3. Frequency Response for Rational System Functions
4. Relationship Between Magnitude and Phase
5. All-Pass and Minimum-phase system
6. Linear Systems with Generalized Linear Phase

- If a **stable** LTI system has a rational system function

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

- Its frequency response is

$$H(e^{j\omega}) = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{\sum_{k=0}^N a_k e^{-j\omega k}}$$

$$H(e^{j\omega}) = \left(\frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k e^{-j\omega})}{\prod_{k=1}^N (1 - d_k e^{-j\omega})} \quad |H(e^{j\omega})| = \left| \left(\frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M |1 - c_k e^{-j\omega}|}{\prod_{k=1}^N |1 - d_k e^{-j\omega}|} \right|$$

$$|H(e^{j\omega})|^2 = H(e^{j\omega}) H^*(e^{j\omega})$$

$$|H(e^{j\omega})|^2 = \left(\frac{b_0}{a_0} \right)^2 \frac{\prod_{k=1}^M (1 - c_k e^{-j\omega})(1 - c_k^* e^{j\omega})}{\prod_{k=1}^N (1 - d_k e^{-j\omega})(1 - d_k^* e^{j\omega})}$$

- Log magnitude

$$\left| H(e^{j\omega}) \right| = \left| \left(\frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k e^{-j\omega})}{\prod_{k=1}^N (1 - d_k e^{-j\omega})} \right|$$

$$20 \log_{10} \left| H(e^{j\omega}) \right| = \text{log magnitude in dB}$$

$$20 \log_{10} \left| \frac{b_0}{a_0} \right| + \sum_{k=1}^M 20 \log_{10} |1 - c_k e^{-j\omega}| - \sum_{k=1}^N 20 \log_{10} |1 - d_k e^{-j\omega}|$$

$$\text{Gain in dB} = 20 \log_{10} \left| H(e^{j\omega}) \right|$$

$$\text{Attenuation in dB} = -20 \log_{10} \left| H(e^{j\omega}) \right| = -\text{Gain in dB}$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$|Y(e^{j\omega})| = |H(e^{j\omega})| |X(e^{j\omega})|$$

$$20 \log_{10} |Y(e^{j\omega})| = 20 \log_{10} |H(e^{j\omega})| + 20 \log_{10} |X(e^{j\omega})|$$

$$\angle Y(e^{j\omega}) = \angle H(e^{j\omega}) + \angle X(e^{j\omega})$$

•Phase Response

$$H(e^{j\omega}) = \left(\frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k e^{-j\omega})}{\prod_{k=1}^N (1 - d_k e^{-j\omega})}$$

$$\angle H(e^{j\omega}) = \angle \frac{b_0}{a_0} + \sum_{k=1}^M \angle [1 - c_k e^{-j\omega}] - \sum_{k=1}^N \angle [1 - d_k e^{-j\omega}]$$

$$\text{grad} [H(e^{j\omega})] = \sum_{k=1}^N \frac{d}{d\omega} (\arg [1 - d_k e^{-j\omega}]) - \sum_{k=1}^M \frac{d}{d\omega} (\arg [1 - c_k e^{-j\omega}])$$

$$\text{grad} \left[H(e^{jw}) \right] = \sum_{k=1}^N \frac{d}{dw} \left(\arg \left[1 - d_k e^{-jw} \right] \right) - \sum_{k=1}^M \frac{d}{dw} \left(\arg \left[1 - c_k e^{-jw} \right] \right)$$

$$1 - d_k e^{-jw} = 1 - r e^{j\theta} e^{-jw} = 1 - r e^{-j(w-\theta)}$$

$$= 1 - r \cos(w - \theta) + jr \sin(w - \theta)$$

$$\longrightarrow \arg \left[1 - d_k e^{-jw} \right] = \arctan \left[\frac{r \sin(w - \theta)}{1 - r \cos(w - \theta)} \right]$$

$$\frac{d}{dw} \left(\arctan \left[\frac{r \sin(w - \theta)}{1 - r \cos(w - \theta)} \right] \right)$$

$$= - \frac{r^2 - r \cos(w - \theta)}{1 + r^2 - 2r \cos(w - \theta)} = - \frac{|d_k|^2 - \text{Re}\{d_k e^{-jw}\}}{1 + |d_k|^2 - 2\text{Re}\{d_k e^{-jw}\}}$$

$$\frac{d}{dw} \left(\arctan \left[\frac{r \sin(w - \theta)}{1 - r \cos(w - \theta)} \right] \right) = - \frac{|d_k|^2 - \operatorname{Re}\{d_k e^{-jw}\}}{1 + |d_k|^2 - 2\operatorname{Re}\{d_k e^{-jw}\}}$$

$$\operatorname{grad} [H(e^{jw})] = \sum_{k=1}^N \frac{d}{dw} \left(\arg [1 - d_k e^{-jw}] \right) - \sum_{k=1}^M \frac{d}{dw} \left(\arg [1 - c_k e^{-jw}] \right)$$

$$= - \sum_{k=1}^N \frac{|d_k|^2 - \operatorname{Re}\{d_k e^{-jw}\}}{1 + |d_k|^2 - 2\operatorname{Re}\{d_k e^{-jw}\}} + \sum_{k=1}^M \frac{|c_k|^2 - \operatorname{Re}\{c_k e^{-jw}\}}{1 + |c_k|^2 - 2\operatorname{Re}\{c_k e^{-jw}\}}$$

1. Frequency Response of LTI Systems
2. System Functions For Systems Characterized by Linear Constant-coefficient Difference equations
3. Frequency Response for Rational System Functions
4. Relationship Between Magnitude and Phase
5. All-Pass and Minimum-phase system
6. Linear Systems with Generalized Linear Phase

- Two systems with

$$H_1(z) = \frac{2(1 - z^{-1})(1 + 0.5z^{-1})}{(1 - 0.8e^{j\pi/4}z^{-1})(1 - 0.8e^{-j\pi/4}z^{-1})}$$

$$H_2(z) = \frac{(1 - z^{-1})(1 + 2z^{-1})}{(1 - 0.8e^{j\pi/4}z^{-1})(1 - 0.8e^{-j\pi/4}z^{-1})}$$

have same **magnitude-squared** system function

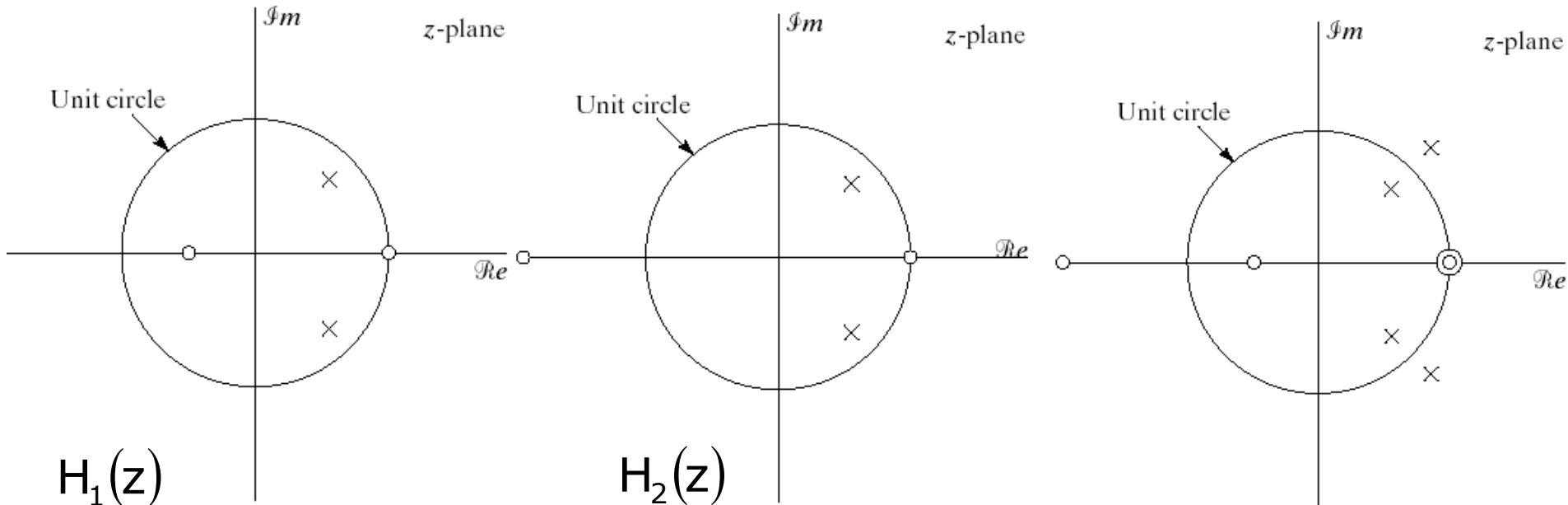
$$2z(1+0.5z^{-1})2z^{-1}(1+0.5z) = (1+2z)(1+2z^{-1})$$

$$C_1(z) = H_1(z)H_1^*(1/z^*)$$

$$= \frac{2(1-z^{-1})(1+0.5z^{-1})2(1-z)(1+0.5z)}{(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{-j\pi/4}z^{-1})(1-0.8e^{-j\pi/4}z)(1-0.8e^{j\pi/4}z)}$$

$$= C_2(z) = H_2(z)H_2^*(1/z^*)$$

$$H_1(z) = \frac{2(1 - z^{-1})(1 + 0.5z^{-1})}{(1 - 0.8e^{j\pi/4}z^{-1})(1 - 0.8e^{-j\pi/4}z^{-1})} \quad H_2(z) = \frac{(1 - z^{-1})(1 + 2z^{-1})}{(1 - 0.8e^{j\pi/4}z^{-1})(1 - 0.8e^{-j\pi/4}z^{-1})}$$

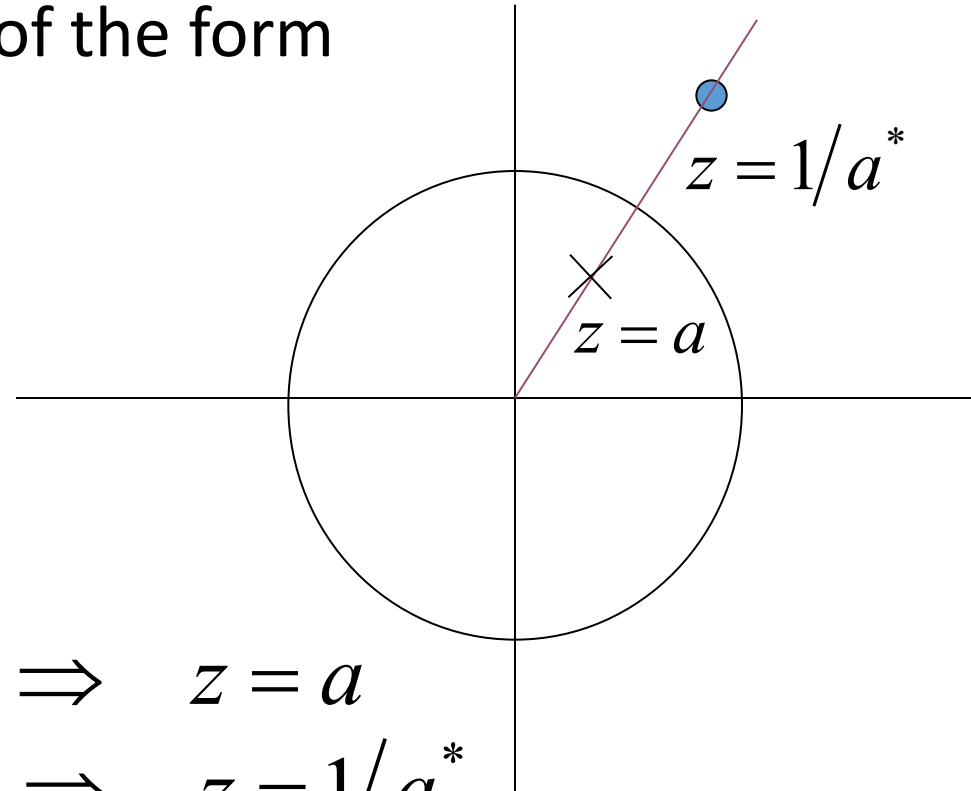


$$C(z) = H_1(z)H_1^*(1/z^*) = H_2(z)H_2^*(1/z^*)$$

1. Frequency Response of LTI Systems
2. System Functions For Systems Characterized by Linear Constant-coefficient Difference equations
3. Frequency Response for Rational System Functions
4. Relationship Between Magnitude and Phase
5. All-Pass and Minimum-phase system
6. Linear Systems with Generalized Linear Phase

- A stable system function of the form

$$H_{ap}(z) = \frac{z^{-1} - a^*}{1 - az^{-1}}$$



$$H_{ap}(z) H_{ap}^*\left(1/z^*\right) = 1$$

- POLE: $1 - az^{-1} = 0 \Rightarrow z = a$
- ZERO: $z^{-1} - a^* = 0 \Rightarrow z = 1/a^*$

- General Form of All-Pass System

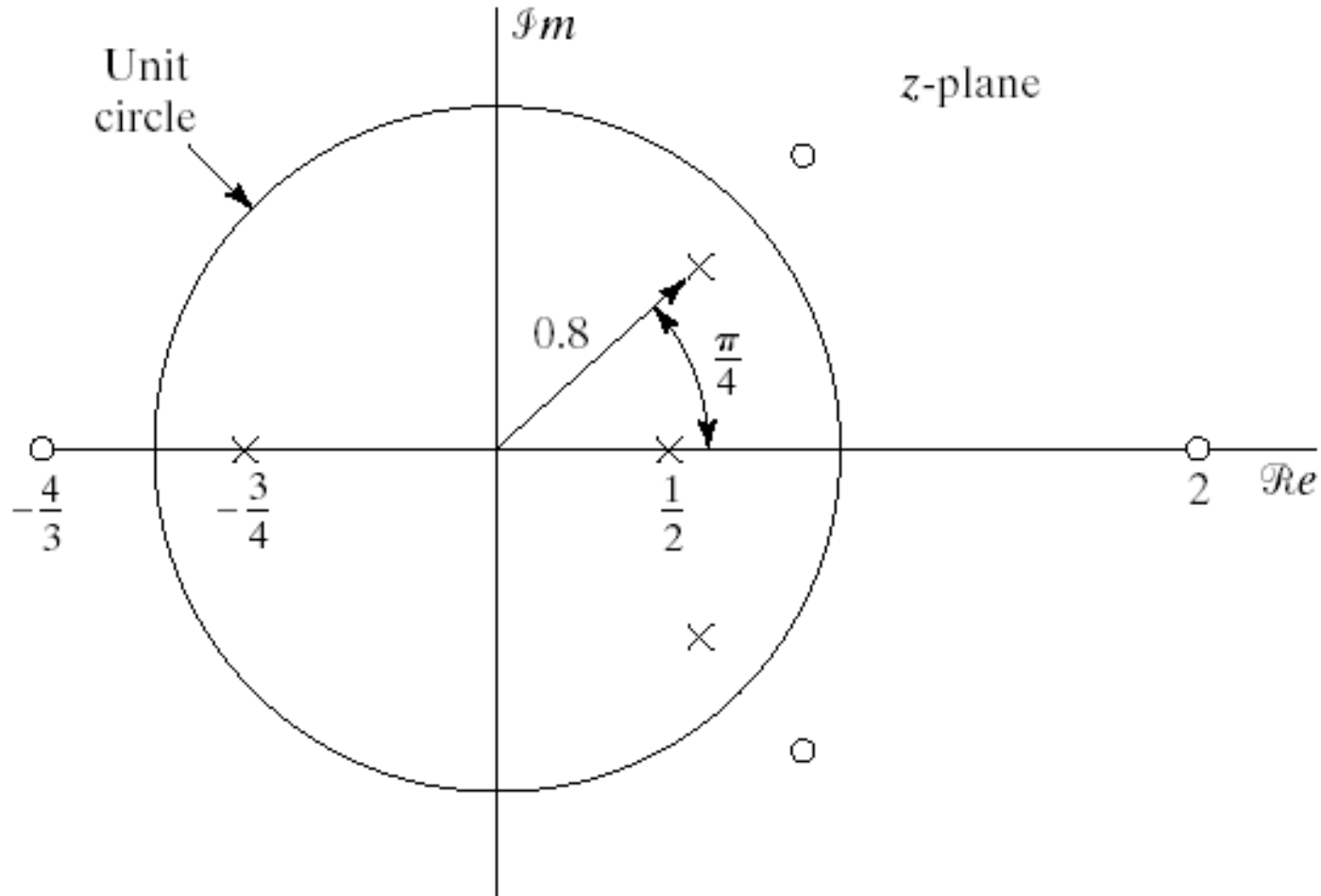
$$H_{ap}(z) = A \prod_{k=1}^{M_r} \frac{z^{-1} - d_k}{1 - d_k z^{-1}} \prod_{k=1}^{M_c} \frac{(z^{-1} - e_k^*)(z^{-1} - e_k)}{(1 - e_k z^{-1})(1 - e_k^* z^{-1})}$$

where A is a positive constant

d_k 's are the real poles of $H_{ap}(z)$

e_k 's are the complex poles of $H_{ap}(z)$

ALL-PASS AND MINIMUM-PHASE SYSTEM



- It has unity magnitude response for all ω

$$H_{ap}(z) = \frac{z^{-1} - a^*}{1 - az^{-1}}$$

$$H_{ap}(e^{j\omega}) = \frac{e^{-j\omega} - a^*}{1 - ae^{-j\omega}} = e^{-j\omega} \frac{1 - a^* e^{j\omega}}{1 - ae^{-j\omega}}$$

$$\left| H_{ap}(e^{j\omega}) \right| = \left| e^{-j\omega} \right| \left| \frac{1 - a^* e^{j\omega}}{1 - ae^{-j\omega}} \right| = 1$$

•Phase response

$$H_{ap}(e^{j\omega}) = \frac{e^{-j\omega} - a^*}{1 - ae^{-j\omega}} = \frac{e^{-j\omega} - re^{-j\theta}}{1 - re^{j\theta}e^{-j\omega}} \quad (\text{if } a = re^{j\theta})$$

$$\begin{aligned} \square \frac{e^{-j\omega} - re^{-j\theta}}{1 - re^{j\theta}e^{-j\omega}} &= \square \frac{e^{-j\omega}(1 - re^{j(\omega-\theta)})}{(1 - re^{-j(\omega-\theta)})} \\ &= \square \frac{e^{-j\omega}(1 - r \cos(\omega - \theta) - jr \sin(\omega - \theta))}{1 - r \cos(\omega - \theta) + jr \sin(\omega - \theta)} \\ &= -\omega - 2 \arctan \left[\frac{r \sin(\omega - \theta)}{1 - r \cos(\omega - \theta)} \right] \end{aligned}$$

- Minimum-Phase Systems
 - For a stable and causal LTI system, all the **poles** must be inside the unit circle.
 - If its inverse system is also stable and causal, all the **zeros** must be inside the unit circle.
 - **Minimum-phase systems**: all the poles and zeros are inside unit circle.

- Minimum-Phase and All-Pass Decomposition
 - Any rational system function can be expressed as
 - Suppose $H(z)$ has one zero outside the unit circle at $z = 1/c^*$, $|c| < 1$, and the remaining poles and zeros are inside the unit circle.

$$H(z) = H_{\min}(z)H_{ap}(z)$$

$$\begin{aligned} H(z) &= H_1(z)(z^{-1} - c^*) \\ &= H_1(z)(1 - cz^{-1}) \frac{(z^{-1} - c^*)}{(1 - cz^{-1})} = H_{\min}(z) \frac{(z^{-1} - c^*)}{(1 - cz^{-1})} \end{aligned}$$

- Example: Minimum-Phase/All-Pass Decomposition

$$H_1(z) = \frac{1 + 3z^{-1}}{1 + \frac{1}{2}z^{-1}} \quad \begin{array}{l} \text{pole : } z = -\frac{1}{2} \text{ inside the unit circle} \\ \text{zero : } z = -3 : \text{outside the unit circle} \end{array}$$

reflect this zero inside the unit circle: $z = -\frac{1}{3}$

$$H_{ap}(z) = \frac{z^{-1} + \frac{1}{3}}{1 + \frac{1}{3}z^{-1}} \quad H_{\min}(z) = 3 \frac{1 + \frac{1}{3}z^{-1}}{1 + \frac{1}{2}z^{-1}}$$

$$H_2(z) = \frac{\left(1 + \frac{3}{2} e^{j\pi/4} z^{-1}\right) \left(1 + \frac{3}{2} e^{-j\pi/4} z^{-1}\right)}{1 - \frac{1}{3} z^{-1}}$$

pole : $z = \frac{1}{3}$ inside the unit circle

zero : $z = -\frac{3}{2} e^{\pm j\pi/4}$: outside the unit circle

reflect this zero inside the unit circle:

$$z = -\frac{2}{3} e^{\pm j\pi/4}$$

$$H_2(z) = \frac{\left(1 + \frac{3}{2}e^{j\pi/4}z^{-1}\right)\left(1 + \frac{3}{2}e^{-j\pi/4}z^{-1}\right)}{1 - \frac{1}{3}z^{-1}} =$$

$$\frac{9}{4} \frac{\left(1 + \frac{2}{3}e^{j\pi/4}z^{-1}\right)\left(1 + \frac{2}{3}e^{-j\pi/4}z^{-1}\right)}{1 - \frac{1}{3}z^{-1}} \times \frac{\left(z^{-1} + \frac{2}{3}e^{-j\pi/4}\right)\left(z^{-1} + \frac{2}{3}e^{j\pi/4}\right)}{\left(1 + \frac{2}{3}e^{j\pi/4}z^{-1}\right)\left(1 + \frac{2}{3}e^{-j\pi/4}z^{-1}\right)}$$

$$= H_{\min}(z) H_{ap}(z)$$

1. Frequency Response of LTI Systems
2. System Functions For Systems Characterized by Linear Constant-coefficient Difference equations
3. Frequency Response for Rational System Functions
4. Relationship Between Magnitude and Phase
5. All-Pass and Minimum-phase system
6. Linear Systems with Generalized Linear Phase

- In designing filters, it's desired to have nearly constant magnitude response and zero phase in bandpass.
- For causal systems, zero phase is not attainable, and some phase distortion must be allowed.

- The effect of linear phase (constant group delay) with integer slope is a simple time shift.
- A nonlinear phase, on the other hand, can have a major effect on the shape of a signal, even when the frequency-response magnitude is constant

- System with Linear Phase

$$H_{id} \left(e^{jw} \right) = e^{-jw\alpha}, \quad |w| < \pi \quad \alpha : \text{real}$$

$$\left| H_{id} \left(e^{jw} \right) \right| = 1 \quad \angle H_{id} \left(e^{jw} \right) = -w\alpha$$

$$\text{grd} \left[H_{id} \left(e^{jw} \right) \right] = \alpha$$

$$h_{id} [n] = \frac{\sin \pi (n - \alpha)}{\pi (n - \alpha)}, \quad -\infty < n < \infty$$

$$= \delta (n - \alpha) \quad \text{if } \alpha \text{ is integer}$$

$$y[n] = x[n] * \frac{\sin \pi (n - \alpha)}{\pi (n - \alpha)} = \sum_{k=-\infty}^{\infty} x[k] \frac{\sin \pi (n - k - \alpha)}{\pi (n - k - \alpha)}$$

•Specially, if $\alpha = n_d : \text{integer}$

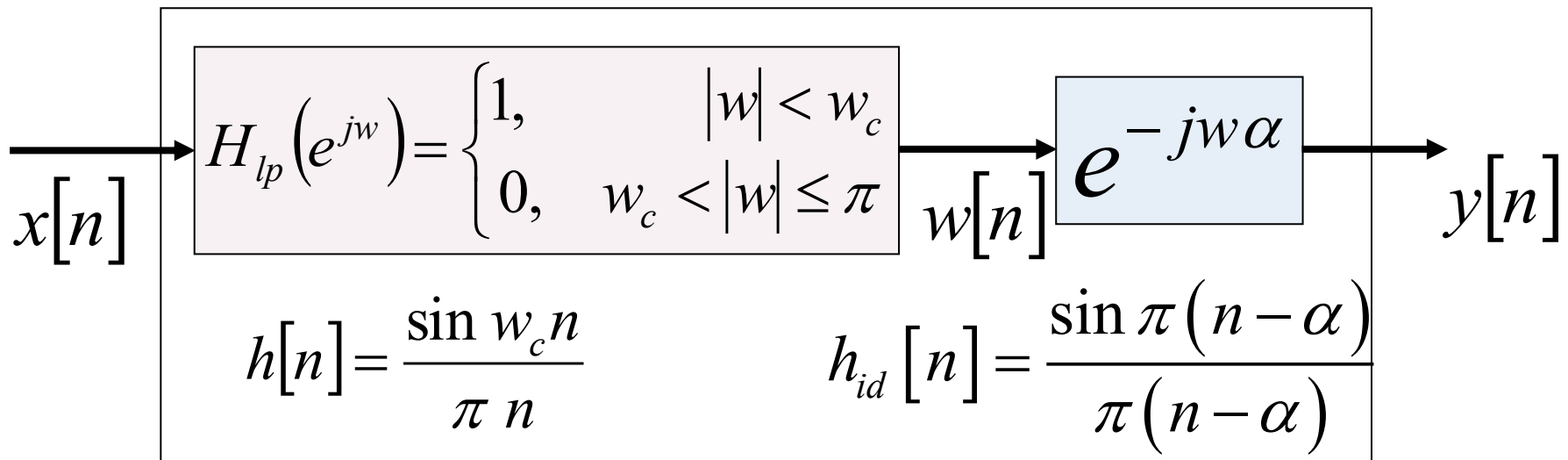
$$h_{id}[n] = \frac{\sin \pi(n - \alpha)}{\pi(n - \alpha)}, \quad -\infty < n < \infty$$

$$h_{id}[n] = \delta[n - n_d]$$

$$y[n] = x[n] * \delta[n - n_d] = x[n - n_d]$$

- Linear-phase ideal lowpass filter

$$H_{lp}(e^{j\omega}) = \begin{cases} e^{-j\omega\alpha}, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases}$$



The corresponding
impulse response is

$$h_{lp}[n] = \frac{\sin \omega_c(n - \alpha)}{\pi(n - \alpha)}$$

- In this chapter we have learnt:
 - Frequency Response of LTI Systems
 - System Functions for Systems Characterized by Linear Constant-coefficient Difference Equations
 - Relationship Between Magnitude and Phase



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

THANK YOU
